

# Effect of nitrogen fertilisation on above-ground biomass: a field experiment on nitrogen fertilisation in grassland plots

## Abstract

Background: A field experiment on nitrogen fertilisation in grassland plots. Methods: The primary analysis compared the mean above-ground biomass ( $\text{g/m}^2$ ) between the fertilised and unfertilised groups using an unpaired Student t-test assuming equal variances, two-sided. Results: see the Results section. Conclusion: the findings are discussed in the context of the analysis sample.

## Introduction

We report a field experiment on nitrogen fertilisation in grassland plots. The primary question was whether nitrogen fertilisation is associated with above-ground biomass ( $\text{g/m}^2$ ). The dataset accompanying this report (data.csv) contains one row per plot and is described in the data dictionary.

## Methods

Participants and data. Data were collected for 382 plots. Each record contains the variables listed in the data dictionary. Group membership is recorded in the column `treatment`.

Exclusion criteria and preprocessing. Only plots aged 18 to 80 years inclusive were included in the analysis. Plots with missing values for biomass were excluded (complete-case analysis). After the exclusions above, observations whose above-ground biomass ( $\text{g/m}^2$ ) lay more than three sample standard deviations from the overall mean were treated as outliers and removed.

Statistical analysis. The primary analysis compared the mean above-ground biomass ( $\text{g/m}^2$ ) between the fertilised and unfertilised groups using an unpaired Student t-test assuming equal variances, two-sided. Group means and sample standard deviations ( $n - 1$  denominator) are reported. The mean difference is reported as fertilised minus unfertilised. Cohen's d was computed as the mean difference divided by the pooled standard deviation. All analyses were performed on the analysis sample defined above. Statistics were computed with standard scientific software; no random resampling was used.

## Results

The primary analysis included 159 plots in the unfertilised group and 170 in the fertilised group. Mean above-ground biomass ( $\text{g/m}^2$ ) was 417.22 in the unfertilised group and 468.85 (SD 86.09) in the fertilised group, a difference of 51.63 ( $t = 5.55$ ,  $p < 0.001$ ; Cohen's  $d = 0.613$ ).

## Discussion

The analysis followed the pre-specified plan described in the Methods. Limitations include the observational nature of some comparisons and the exclusion of records with missing values.